WEEK 08 REPORT

DASHBOARD CONSTRUCTION

Project: Moneyball – IPL Player Valuation
Course: 24ADI204 – Data Science and Visualization
Event: Data-Viz Hack-A-Bit (2026–2027) **Team:** 05

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Submitted as the Week 08 practical documentation for the team project.

1. Executive Summary

Week 08 focused on dashboard construction for the Moneyball – IPL Player Valuation project. A Streamlit application was developed with Plotly visualizations to present the project as a six-chapter data story. The dashboard combines the dataset overview, data-quality findings, transformation comparisons, auction-price analysis, principal component analysis (PCA), and a player valuation model.

The submitted screenshots show that the dashboard is running in a browser through a Cloudflare Quick Tunnel. The interface includes interactive chapter selection, season and role filters, charts, tables and explanatory text.

2. Objectives

• Build an interactive dashboard using Streamlit and Plotly.
• Organize the project findings into a coherent six-chapter narrative.
• Allow users to filter records by IPL season and player role.
• Present data quality, transformation, auction-price and PCA findings visually.
• Demonstrate a Moneyball-style valuation approach using previous-season performance.

3. Tools and Technologies

Python: dashboard programming language.
Streamlit: interactive web application framework.
Plotly: interactive charts and visualizations.
Pandas and NumPy: data loading, filtering and numerical operations.
Scikit-learn: StandardScaler, PCA and LinearRegression.
Google Colab: notebook execution environment.
Cloudflare Quick Tunnel: temporary public access to the locally running Streamlit application.

4. Dashboard Structure

Chapter 1 – Context

The opening chapter introduces the question: does IPL auction spending reward player performance? The full-dataset screenshot reports 3,311 player-season records, 794 unique player values, 369,972 total runs and 13,113 total wickets. It also shows average runs by season and the distribution of players by role.

Chapter 2 – Data Quality Report

The dashboard reports a dataset size of 3,311 rows × 27 columns, 13,099 missing cells and zero duplicate rows. A missing-value percentage chart and a column-level summary table help identify fields that still contain missing values.

Chapter 3 – Fixing the Data (Before vs After)

The user can select a performance column and compare its raw distribution with a log1p-transformed distribution and a standardized version. The screenshot for runs displays the raw skewness as 1.416 and the log1p skewness as -0.324. The table and three distribution plots provide visual evidence of the transformation.

Chapter 4 – The Conflict

This chapter compares final auction price with a selectable performance metric, such as runs, wickets, sixes or strike rate. The supplied screenshot shows final price versus runs, with points coloured by player role. The dashboard also provides price-spread visualizations and a correlation view. The story highlights that auction price does not rise in a simple, perfectly linear way with a single performance statistic.

Chapter 5 – Hidden Structure (PCA)

The PCA chapter standardizes selected performance features, calculates explained variance and plots players in the first two principal-component dimensions. The screenshot shows the explained-variance chart and player scatter plot. The displayed dashboard message indicates that two components retain about 90% of the variance for the current

data and selected feature set.

Chapter 6 – Resolution (Moneyball Value Model)

The final chapter explains that a franchise must estimate a player's value before the upcoming season, so the model uses previous-season performance features and season information. It compares the predicted expected price with the actual final price. A negative gap is labelled as priced below expectation, while a positive gap is labelled as priced above expectation. The supplied screenshot reports an R^2 value of 0.32 for 1,464 priced players with previous-season history under the current filters.

5. Key Observations

• The full-data context view shows 3,311 player-season records and 794 distinct player values.
• The Week 3 cleaned dataset is represented in the dashboard as 3,311 × 27, with 13,099 remaining missing cells and no duplicate rows.
• The runs transformation example reduces skewness from 1.416 to -0.324 after log1p.
• The auction-price scatter plot indicates substantial variation in price at similar performance levels.
• The PCA view communicates how a smaller number of components can summarize much of the variation in the selected performance features.
• The valuation model's displayed R^2 of 0.32 suggests that previous-season performance alone does not explain all variation in auction prices.

6. Methodology and Implementation

The uploaded Week 08 notebook installs the required libraries, accepts a CSV upload, writes the Streamlit source file as app.py, and starts the Streamlit server on port 8501. It then starts the Cloudflare tunnel to expose the application through a temporary public URL. The dashboard source code implements the six chapters and the interactive filters and visualizations described above.

7. Important Limitations

Valuation model evaluation: In the uploaded notebook's Chapter 6 code, LinearRegression is fitted on Xa and the displayed R^2 is calculated using model.score(Xa, ya) on that same data. This is an in-sample training score, not a held-out test score. It should not be presented as proof of future predictive accuracy. A train/test split or cross-validation is recommended for a stronger evaluation.
Filter dependence: Dashboard values and charts change when seasons or roles are selected. Screenshots of filtered views should be labelled as filtered results. The overall context screenshot is the full-data view.
Temporary hosting: Cloudflare Quick Tunnel URLs are temporary and may stop working when the Colab runtime or tunnel process stops. The notebook and app.py source should be retained for reproducibility.

8. Conclusion

The Week 08 work converts the project's analysis into an interactive dashboard with a clear narrative from context and data quality to transformations, auction-price analysis, PCA and player valuation. The screenshots demonstrate the six chapters and the interactive visual components. The dashboard provides a useful way to explore the relationship between IPL player performance and auction price, while the valuation results should be interpreted cautiously because the current R^2 is calculated on the training data.

9. Submission Files

Recommended Week 08 submission package:
1. Week08_Dashboard_Construction.ipynb – executed Google Colab notebook.
2. app.py – Streamlit dashboard source code.
3. Week08_Dashboard_Construction_Report_Team05.pdf – this report.
4. Dashboard screenshots – visual evidence for all six chapters.
5. The cleaned dataset used by the dashboard, if required by the instructor.

The DSV rulebook describes Week 08 as dashboard construction using Streamlit/Plotly and requires the final project GitHub deliverable to include the project report, source code, raw and cleaned datasets, requirements.txt and an optional demo video. Follow the faculty's specific weekly submission instructions for the Week 08 checkpoint.

10. Dashboard Screenshot Evidence

The following screenshots were supplied with the Week 08 report request. They are included as evidence of the dashboard's visible chapters and charts. Values shown in filtered screenshots depend on the active season and role selections.

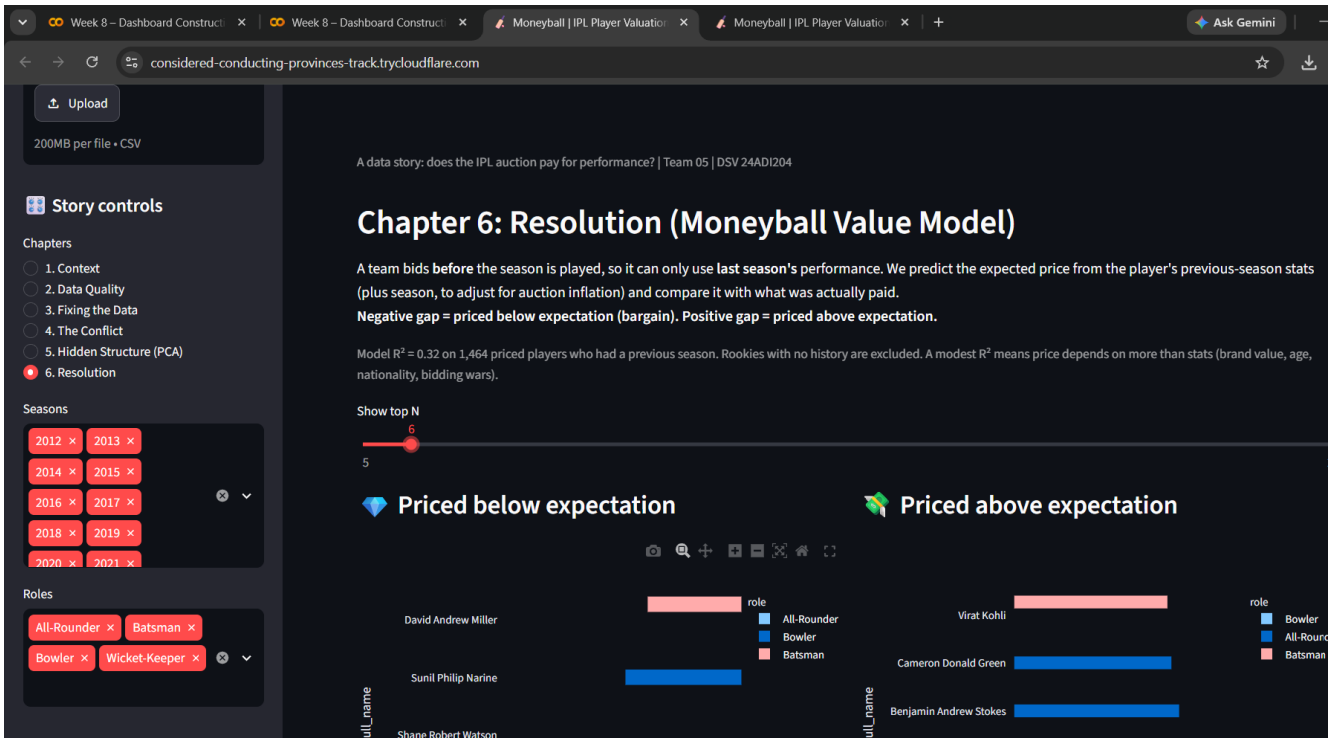


Figure 1. Chapter 1 – Context: full dataset summary and player-role distribution.

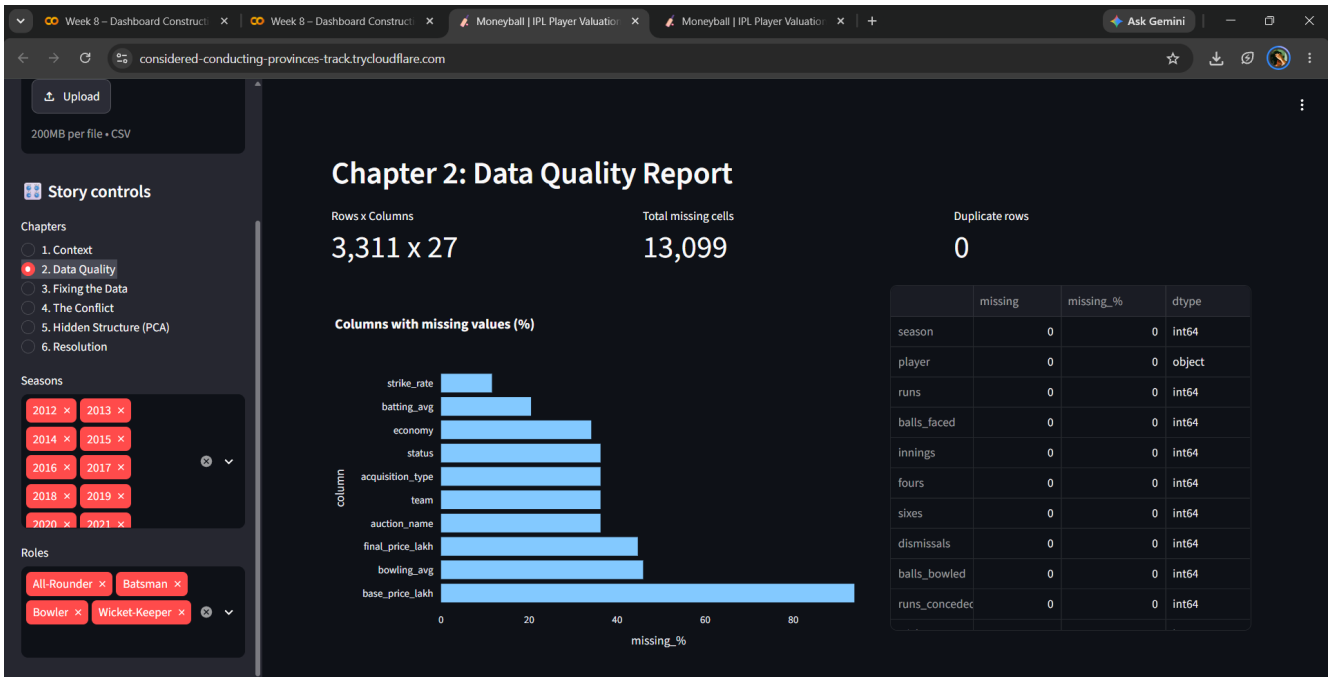


Figure 2. Chapter 2 – Data Quality: dimensions, missing cells, duplicates and missing-value chart.

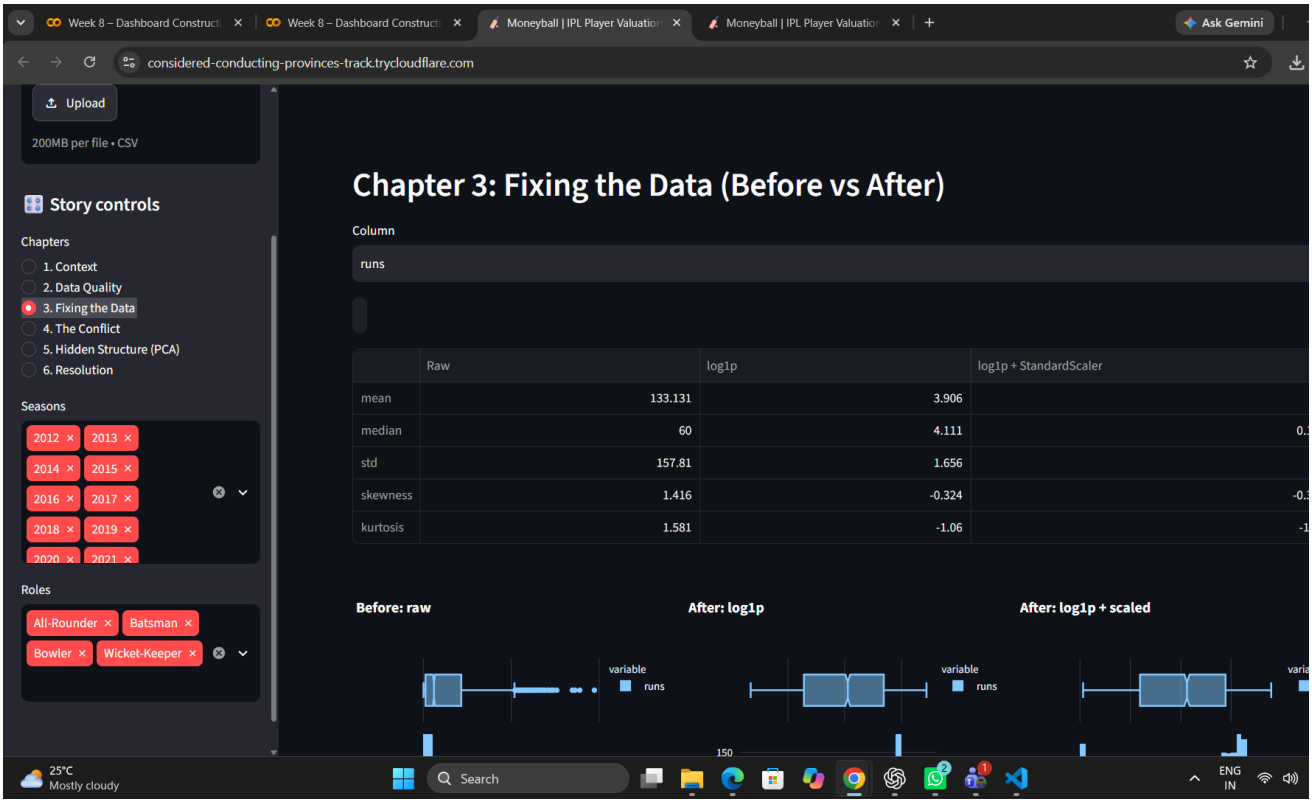


Figure 3. Chapter 3 – Fixing the Data: raw, log1p and standardized distributions.

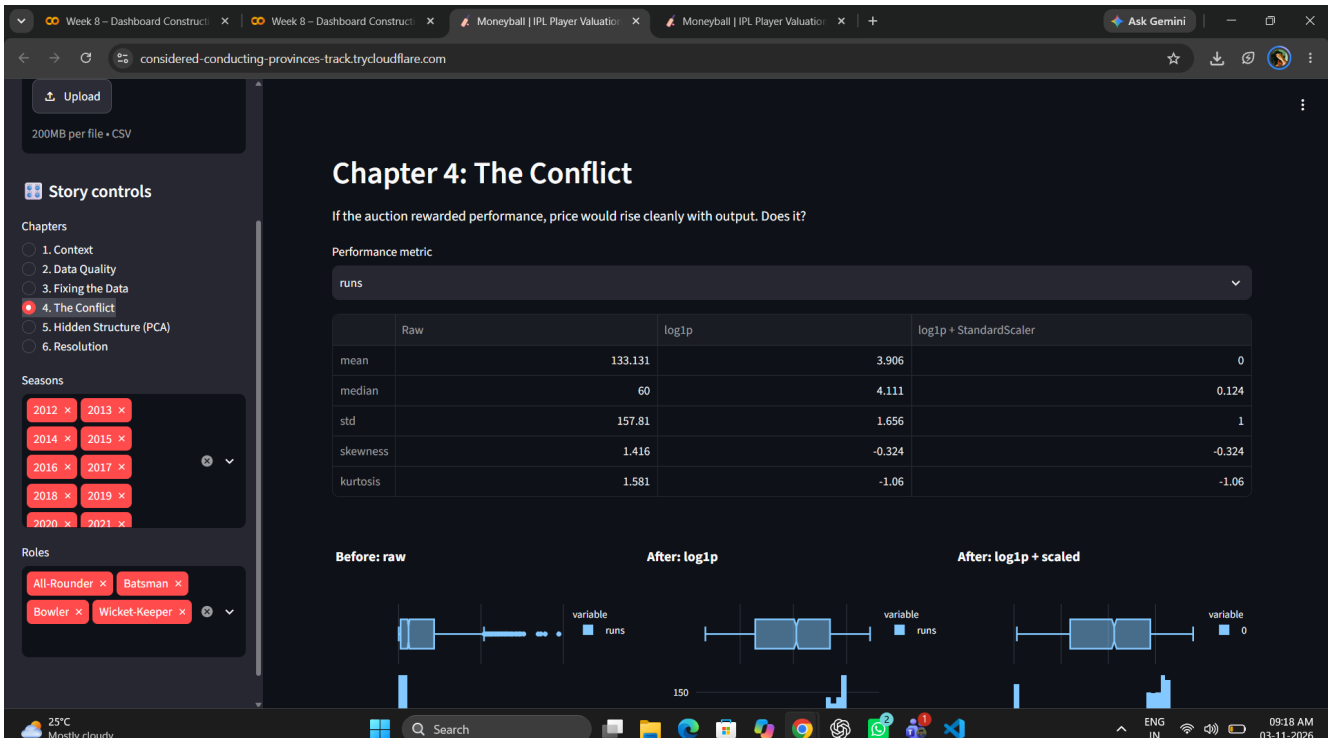


Figure 4. Chapter 4 – The Conflict: final auction price versus runs.



Figure 5. Chapter 4 – Transformation comparison and descriptive statistics.

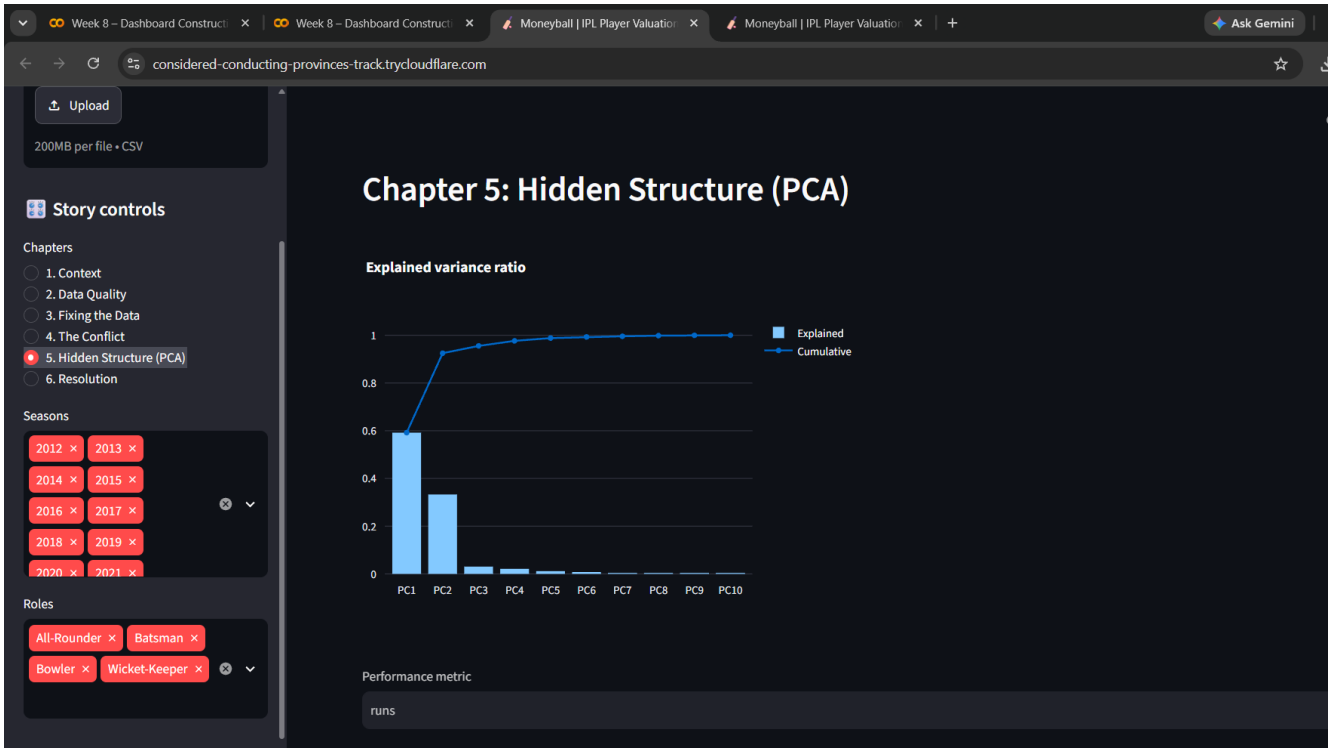


Figure 6. Chapter 5 – PCA explained variance and player positions in PC space.

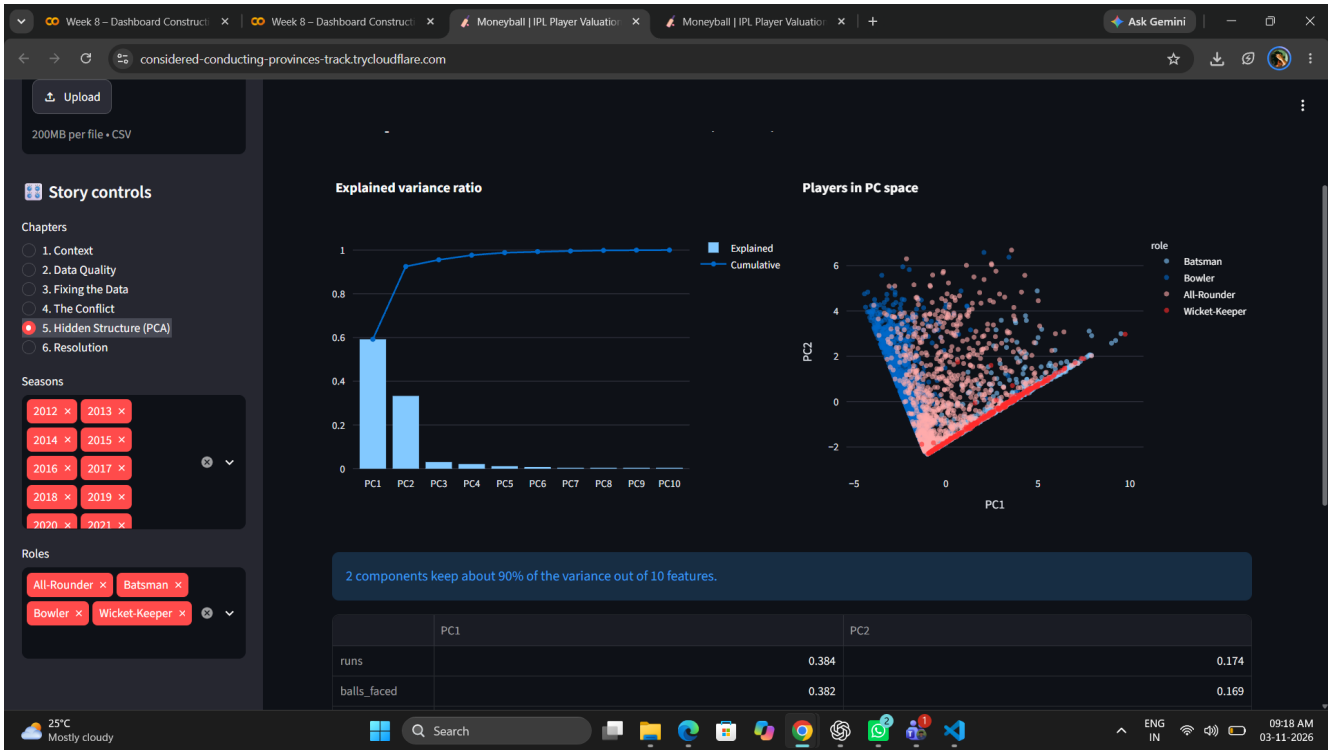


Figure 7. Chapter 5 – PCA explained variance chart and feature view.

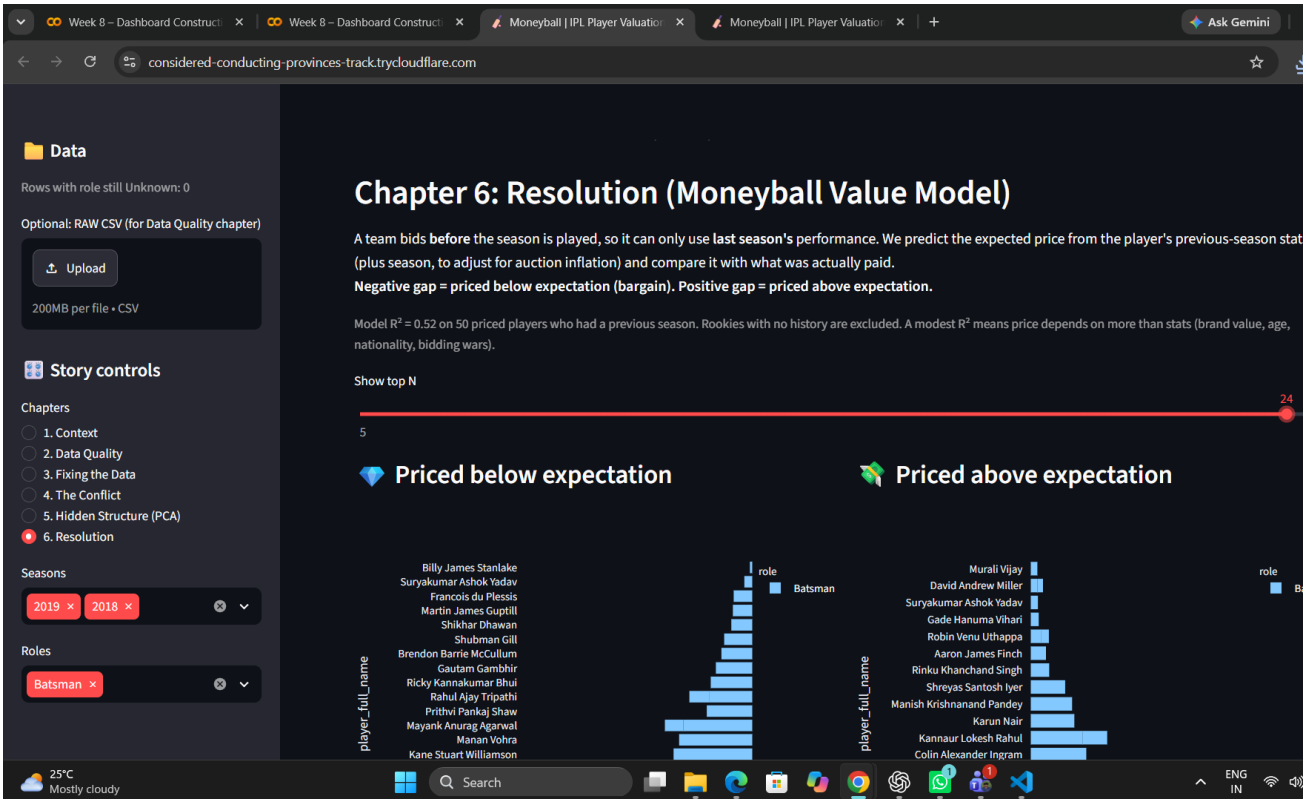


Figure 8. Chapter 6 – Moneyball value model: players priced below and above expectation.